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Recovery of elemental sulphur from anaerobic effluents through the biological oxidation of sulphides

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ABSTRACT

The aim of the present study was to evaluate the biological oxidation of sulphide in two different UASB reactors by assessing the occurrence of oxidized forms of sulphur in the effluents and the amount of S^0 that could be recovered in the process. The bioreactors employed were an anaerobic hybrid (AH) reactor employing porous *polyurethane foam* as support media and a micro-aerated UASB reactor equipped with an aeration device above the digestion zone. The AH reactor produced a final effluent containing low concentrations of S^{2-} (3.87% of total sulphur load). It was achieved due to a complete oxidation of 56.1% of total sulphur. The partial biological oxidation that occurred in the AH reactor allowed the recovery of 30% of the sulphur load as S^0. The effluent from the micro-aerated UASB reactor contained 5% of the sulphur load in the form of S^{2-}, while 20.9% was present as dissolved SO_4^{2-} and 46% was precipitated as S^0. It is concluded that the AH reactor or micro-aeration carried out above the digestion zone of the UASB reactor favoured the biological oxidation of S^{2-} and the release of odourless effluents. Both technologies represent feasible and low-cost alternatives for the anaerobic treatment of domestic sewage.

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Anaerobic system; sulphide removal; elemental sulphur recovery; odourless effluents; micro-aeration

1. Introduction

The upflow anaerobic sludge blanket (UASB) reactor is an attractive technology for the treatment of wastewaters despite various limitations, which include the release of gaseous emissions containing H_2S.[1–3] This highly toxic sulphide is produced when *sulphate*-reducing bacteria present in the sludge bed obtain energy by oxidizing organic compounds while reducing SO_4^{2-} to H_2S through dissimilatory sulphate reduction.[4] Moreover, the process of sulphidogenesis is harmful to methanogenic archaea, and leads to a reduction in the removal of chemical oxygen demand (COD) and a diminution in the concentration of methane in the biogas.[5–10]

The type of sulphide present in liquid waste during biological treatment depends on pH, with volatile H_2S being the predominant form in the acid medium while HS^- and S^{2-} are the principal species in neutral and alkaline media, respectively.[11] Partial biological oxidation of sulphide can occur in anaerobic, anoxic or aerobic environments in which the electron acceptors are CO_2, NO_3^- or O_2, respectively.[7] In anaerobic environments, *anoxygenic phototrophic bacteria, such as purple and green sulphur bacteria,* use energy from light to reduce CO_2 by transferring electrons from elemental sulphur (S^0).While purple bacteria store intracellular sulphur particles inside the cell and carry out complete oxidation to SO_4^{2-}, green bacteria deposit sulphur granules outside the cell and perform incomplete oxidation.[12] It is noteworthy, however, that partial as well as complete oxidation of sulphides can occur with both groups of bacteria. In this context, Garcia et al. [2] reported that cyanobacteria (*Phormidium* and *Pseudoanabaena*) and sulphur-oxidizing bacteria similar to *Beggiatoa* or *Thiotrix*, present in the scum layer of the settler compartment of a UASB reactor, reduced sulphide concentration sand could play a role in minimizing odour emission.

The supply of oxygen at low concentrations may remove the main product of sulphide oxidation, that is, insoluble elemental sulphur, from the liquid phase.[13] The presence of small quantities of oxygen, about 0.1 $mgO_2\,L^{-1}$, in anaerobic reactors is sufficient to create micro-aerobic conditions that enable microbial communities to partially oxidize sulphide to S^0, according to Equation (1):[13–17]

$$HS^- + {}^1/_2O_2 \rightarrow S^0 + OH^-. \quad (1)$$

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Van der Zee et al. [18] studied the fate of sulphur compounds and oxygen during micro-aerobic sulphide oxidation and found that the introduction of a low air flow of 0.7–0.9 $m^3\ m^{-3}$ reactor volume day^{-1} (corresponding to an O_2/S molar ratio of 8–10) to a fluidized bed reactor was sufficient to reduce the H_2S content in the biogas to levels that were almost undetectable (~0.02%).[5]

Application of micro-aerobic UASB as an alternative technology would not only assist in solving the problem of odour emission but would also allow the recovery of S^0 present in sewage. The elemental sulphur so recovered could be employed in the industrial production of H_2SO_4 or $(NH_4)_2SO_4$ for use in agriculture, thus adding extra value to the treatment process.[19,20]

Considering the problems of sulphide formation in anaerobic reactors, the present study aimed (i) to evaluate the performance of two different UASB reactors in the treatment of domestic wastewater by assessing the occurrence of oxidized forms of sulphur (S^0 and SO_4^{2-}) in the effluents and (ii) to estimate the amount of S^0 recovered in the processes. The bioreactors employed were an anaerobic hybrid (AH) reactor comprising porous *polyurethane foam* as support media and a micro-aerated UASB reactor equipped with an aeration device above the digestion zone.

2. Materials and methods

2.1. Site of study and origin of sewage

The experiments were carried out at the Estação Experimental de Tratamento Biológico de Esgotos Sanitários (EXTRABES) located in an area belonging to the Companhia de Água e Esgotos da Paraíba (CAGEPA) and the Universidade Estadual da Paraiba, Campina Grande, PB, Brazil (7°14′24″ S; 35°53′02″ W; 550 m altitude). The municipal wastewaters employed in the experiments were collected using a submersible pump installed in an inspection well of the CAGEPA east sewer interceptor pipe. The sewage flowed by gravity through 2 cm diameter plastic tubing to an *equalization tank* (100 L) equipped with a peristaltic metering pump with preset flow system to feed the reactor.

2.2. Characteristics and operational conditions of the anaerobic reactors

The AH reactor was constructed from polyvinyl chloride (PVC) tubing (2 m × 20 cm diameter) and comprised a UASB unit of useful volume 27 L operating with a hydraulic retention time (HRT) of 6 h, and an anaerobic filter of useful volume 25 L operating with an HRT of 5:30 h. The anaerobic filter support consisted of polyurethane sponge cubes (3 × 3 × 3 cm) that occupied 70 cm of the filter height and presented 97% void volume. The raw sewage (influent) entered the UASB unit at the base and moved with a continuous ascendant flow provided by the peristaltic pump. A tap was provided at the top of the UASB unit for the collection of effluent samples, and the final treated effluent was collected with the aid of another tap located above the anaerobic filter. The growth of dispersed and adherent sludge in the UASB and filter units, respectively, could be monitored in samples collected using further 10 taps spaced throughout the reactor body. A schematic of the AH system projected for a bench-scale experiment is presented in Figure 1(a), and the physical and operational parameters are given in Table 1. The complete reactor presented a useful volume of 52 L, operated with a total HRT of 11:30 h and had a sewage treatment capacity of 108 L day^{-1}. The inoculum used in starting the system was anaerobic sewage (6.5 L) with biomass containing 48 g L^{-1} of total solids and 28 g L^{-1} of total volatile solids. The reactor was monitored over a six-month operational period. During this period samples of 100 mL of the effluents were collected weekly (24 samples) on Wednesdays at 8 a.m. in both reactors. In the micro-aerated UASB reactor, monthly samples of sludge were collected (500 mL) for determination of sulphur species.

The micro-aerated UASB reactor (Figure 1(b)) was constructed from PVC tubing (2 m × 20 cm diameter) and presented a useful volume of 25 L, of which 17 L (68%) was micro-aerated and 8 L (32%) corresponded to the digestion zone. Sewage entered at the base of the reactor, and the treatment capacity of the reactor was 50 L day^{-1}. The rationale behind the reactor operation was to supply a low flow of oxygen to permit the oxidation of sulphur species produced by sulphate-reducing bacteria during sulphidogenesis, thereby yielding an odourless effluent. Hence, a micro-aeration module with 3.5 L useful volume was connected to the main body of the reactor at point 1 (Figure 1(b)) with the base of the unit located 20 cm below the effluent outlet of the main reactor. The module contained a porous stone sparger and was connected to a nebulizer (Nebular Plus®, Daru, Inhauma, RJ, Brazil) so that a fraction of the UASB effluent could be aerated and returned through a PVC tube (4 mm diameter) that entered the main reactor at point 2 (Figure 1(b)) located 46 cm above the reactor base. The airflow injected daily was monitored using a gas meter (Lao Indústria, Osasco, SP, Brazil) connected to a flow meter (Oxigel, São Paulo, SP, Brazil) measuring in the range of 0–1.5 L min^{-1}. The concentration of dissolved

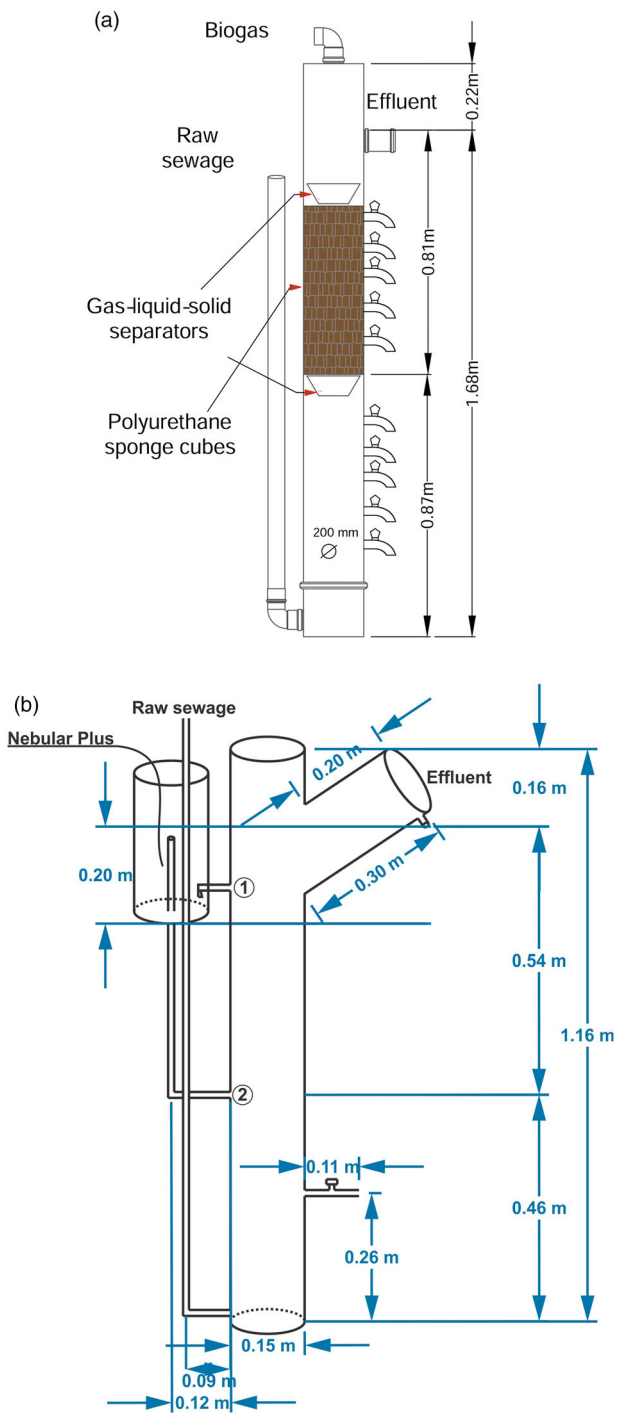


Figure 1. Schematic of the experimental **(a)** anaerobic filter reactor and **(b)** micro-aerated UASB reactor.

oxygen in the micro-aeration module was determined daily with the aid of an Oakton Instruments (Vernon Hills, IL, USA) model DO 6 economy handheld dissolved oxygen meter. The level of dissolved oxygen was verified on a weekly basis using a Beluga-type biogas respirometer (developed at the Universidade Federal da Paraiba) attached to a YSI (Yellow Springs, OH, USA) electrode and the readings were recorded and plotted

Table 1. Physical and operational parameters of the AH and micro-aerated UASB reactors.

Parameter	AH reactor	Micro-aerated UASB reactor
Diameter (mm)	200	150
Material	PVC	PVC
Total height (m)	1.90	1.16
Useful height (m)	1.68	1.00
Total volume (L)	60	20
Useful volume (L)	52	18
HRT of UASB unit (h)	6	12
HRT of anaerobic filter (h)	5.5	–
Influent flow (L day^{-1})	108	50

Notes: HRT, hydraulic retention time; PVC, polyvinyl chloride.

in spreadsheet form.[21] The reactor was monitored over a three-month operational period.

2.3. Analysis of raw sewage and reactor effluents

The standard methods described by the APHA/AWWA/WEF [22] were employed to assess COD (titrimetric closed reflux; APHA/AWWA/WEF section 5220C), sulphide (methylene blue method; APHA/AWWA/WEF section 4500D) and pH (potentiometry). Alkalinity (mg $CaCO_3L^{-1}$) was determined using the Kapp method according to Buchauer,[23] while elemental sulphur was measured colorimetrically using the method of Bartlett and Skoog.[24] The concentrations of SO_4^{2-} present were analysed by ion chromatography using a Dionex™ model ICS-1100 chromatograph with suppressed and detector conductivity (Thermo Scientific, Waltham, MA, USA). The mobile phase was carbonate 4.5 and 0.8 mM bicarbonate. The column (2 × 250 mm) with a particle size of 6.0 µm, capacity column80 µeq/column functional group alkyl/alkanol quaternary ammonium, hydrophobicity low was used. Effluent samples were submitted to two consecutive filtrations through cellulose acetate membranes (0.45 and 0.22 µm) and then diluted five times prior to analysis. Scanning electron microscopy with energy-dispersive X-ray (SEM/EDX) was performed using a TM-1000 bench model instrument (Hitachi, Tokyo, Japan) to observe submicron-scale features of the solid material precipitated in the anaerobic filter effluent and to provide elemental identification and quantitative compositional information. Samples were analysed without prior application of conductive metal coating at a magnification of up 10,000×, depth of focus of 1 mm, resolution of 30 nm, acceleration voltage of 15 kV and low vacuum with variable pressure between 1 and 270 Pa. X-ray diffraction (XRD) analysis of the samples was performed using an XRD-7000 diffractometer (Shimadzu, Kyoto, Japan) and scanned in the Bragg-Brentano mode with a scattering angle 2θ of 5–80°, geometry of θ-2θ, Cu Kα radiation (λ = 1.54 Å), step size 0.02° (2θ steps) , and interval of 0.5 s.

2.4. Mass balance of sulphur species produced by the micro-aerated UASB reactor

The mass balance of sulphur species was calculated from:

$$MS_I = Q_I(I_{SO_4^{2-}} + I_{S^{2-}}), \quad (2)$$

$$MS_E = Q_I(E_{SO_4^{2-}} + E_S^{2-} + E_S^0), \quad (3)$$

$$MS_X = Q_I(X_{SO_4^{2-}} + X_S^{2-} + X_S^0), \quad (4)$$

$$MS_L = MS_I - (MS_E + MS_X), \quad (5)$$

where MS_I, MS_E and MS_X are, respectively, the total loads (mg day^{-1}) of sulphur species in the influent, effluent and sludge, MS_L is the loss (mg day^{-1}) of sulphur in the form of H_2S, and Q_I is the flow rate of influent (L day^{-1}). The concentrations of SO_4^{2-}, S^{2-} and S^0 prefixed with *I*, *E* and *X* relate to the levels (mg L^{-1}) of sulphur species established, respectively, in the influent, effluent and sludge. The mass balances of the sulphur species in the various reactor fractions, namely $MS_{I_{SO_4^{2-}}}$, $MS_{I_S^{2-}}$, $MS_{E_S^{2-}}$, $MS_{E_S^{2-}}$, $MS_{E_S^0}$, $MS_{X_{SO_4^{2-}}}$, $MS_{X_S^{2-}}$ and $MS_{X_S^0}$ were determined from Equations (2)–(4), while the sulphur lost as H_2S could be established from Equation (5).

3. Results and discussion

3.1. Performance of the AH reactor

The chemical characteristics of the raw sewage supplied to the AH reactor and effluents derived from the UASB and the anaerobic filter are given in Table 2. The mean pH values of raw sewage and UASB effluents varied between 7.7 and 7.4 with a standard deviation ≤ 0.30, which is considered substantial taking into account the influence of pH on the competition between sulphate-reducing bacteria and methanogenic archaea. Visser et al. [25] demonstrated that the efficiency of sulphate removal by a UASB reactor is a function of the pH of the effluent, in that the reduction of SO_4^{2-} predominates when the pH is higher than 8 but the production of methane and the reduction of SO_4^{2-} are in equilibrium when the pH varies between 6.75 and 7.5. In the present study, it is likely that the high alkalinity values (determined as mg $CaCO_3$ L^{-1}) of the raw sewage and the reactor effluents enhanced the buffering efficiency of the system and accounted for the relatively small variations in pH.

The concentrations of S^{2-} and SO_4^{2-} in the raw sewage fed to the AH reactor were, respectively, 7.77 and 14.43 mg L^{-1} corresponding to a total sulphur load in the influent of 22.2 mg L^{-1}, whereas the total concentration of sulphur in the UASB effluent was 20.59 mg L^{-1} indicating a loss of approximately 1.61 mg L^{-1} (Table 2). In contrast, the total concentration of the studied anions in the effluent from the anaerobic filter was around 13.32 mg L^{-1}, and they were present mainly in the form of SO_4^{2-} (12.46 mg L^{-1}) rather than S^{2-}(0.86 mg L^{-1}). During the experiment, the top of the anaerobic filter was open to the air, thus permitting gaseous exchange and the ingress of light. Under these conditions, the S^{2-} species produced in the UASB unit were oxidized to SO_4^{2-} and S^0, as has been observed in previous studies.[2,7,26,27]

The marked increase in the concentration of S^{2-} in the UASB effluent compared with that in the raw sewage was probably due to containment of the influent in the equalization tank, which promoted the occurrence of sulphidogenesis prior to entry into the UASB reactor. Indeed, the concentration of S^{2-} in the UASB effluent remained high during the whole experiment, demonstrating the efficiency of sulphate-reducing bacteria in transforming SO_4^{2-} to S^{2-}. Oxidation of S^{2-} in the anaerobic filter unit was mediated by suspended cells of photoautotrophic and/or photoheterotrophic bacteria containing chlorophyll and carotenoid pigments. These organisms were able to survive in such an environment because the top of the filter unit was open to the air and exposed to light.[27,28] As a consequence of this activity, the effluent from the anaerobic filter exhibited a low concentration of S^{2-} (0.86 mg L^{-1}) and the unpleasant odour that is typical of anaerobic effluents was substantially reduced. On the other hand, the concentration of SO_4^{2-} remained high, thus maintaining the complete oxidation of S^{2-} to SO_4^{2-}. It is likely that the population of purple bacteria predominated under the prevailing experimental conditions, thereby favouring complete oxidation to SO_4^{2-}.[12]

Table 2. Chemical profile of the raw sewage fed to the AH reactor and of the effluents derived therefrom.

	Mean ± standard deviation		
Parameter	Raw sewage	Effluent from UASB	Effluent from anaerobic filter
pH	7.7 ± 0.28	7.4 ± 0.21	7.9 ± 0.30
Alkalinity (mg$CaCO_3$ L^{-1})	380 ± 120	420 ± 108	400 ± 98
Raw COD (mgO_2 L^{-1})	400 ± 112	205 ± 88	150 ± 67
SO_4^{2-} (mgS L^{-1})	14.43 ± 2.2	1.62 ± 0.31	12.46 ± 3.8
S^{2-}(mgS L^{-1})	7.77 ± 1.5	18.97 ± 3.9	0.86 ± 0.32
S^0 (mgS L^{-1})	NA	NA	6.66 ± 0.68

Notes: COD, chemical oxygen demand; NA, not applicable; UASB, upflow anaerobic sludge blanket.

The quantitative analysis revealed that 30% (6.66 mg L^{-1}) of the sulphur load present in the raw sewage was recovered as elemental sulphur, while the XRD pattern of the solid material that precipitated from the anaerobic filter effluent (Figure 2(a)) confirmed the predominant presence of S^0. Figure 2(b) presents a SEM image of the surface of this solid material at 1200 × amplification, while the energy-dispersive X-ray (EDX) spectrum, shown in Figure 2(c), displayed only the characteristic peak of elemental sulphur, thereby confirming that 98% of the precipitate present in the anaerobic filter effluent was actually S^0. Silva et al. [29] also reported that areas of a UASB reactor exposed to light supported the growth of bacterial populations employing S^{2-} as electron donor in energy-producing processes and forming insoluble precipitates containing 87% of S^0. In the present study, the total concentration of sulphur in the influent was 22.2 mg L−1, whereas the final anaerobic effluent contained 19.98 mg L^{-1} of the element when all of the chemical species analysed (i.e. SO_4^{2-}, S^{2-} and S^0) were taken into account. It was not possible to determine the fate of the residual portion of the sulphur load, but it was probably discharged in the volatile form (H_2S) or remained undetected in the sediment.

3.2. Performance of the micro-aerated UASB reactor

The chemical characteristics of the raw sewage supplied to the micro-aerated UASB reactor and the effluent derived the reform are given in Table 3. The mean pH values of raw sewage and effluent (7.91 and 7.75, respectively), although slightly above the preferred range of 7.0–7.5, were acceptable given that values below 6.5 and above 8.5 are inhibitory to methanogenic archaea.[30] However, it is important to maintain pH at around 7.0 when attempting to remove sulphides, since in this region 50% of the sulphur species are dissociated (HS^-) and 50% are in the non-dissociated and more toxic form (H_2S).[31] As for the AH reactor, the high alkalinity values of the raw sewage and the UASB effluent enhanced the buffering capacity of the system, resulting in minimal variations in pH.

Table 3. Chemical profile of the raw sewage fed to the micro-aerated UASB reactor and of the effluent derived therefrom.

	Mean ± standard deviation	
Parameter	Raw sewage	UASB effluent
pH	7.91 ± 0.23	7.75 ± 0.12
Alkalinity (mg $CaCO_3$ L^{-1})	373 ± 42.5	416 ± 35.31
Raw COD (mgO_2 L^{-1})	728 ± 165	221 ± 44
SO_4^{2-} (mgS L^{-1})	7.68 ± 1.79	2.10 ± 1.34
S^2 (mgS L^{-1})	2.36 ± 2.43	0.50 ± 0.32
S^0 (mgS L^{-1})	NA	4.62 ± 0.70

Notes: COD, chemical oxygen demand; NA, not applicable.

The concentrations of S^{2-} and SO_4^{2-} in the raw sewage fed to the micro-aerated UASB reactor were 2.36 and 7.68 mg L^{-1}, respectively, corresponding to a total concentration of sulphur in the influent of 10.04 mg L^{-1}(Table 3). In the UASB effluent, the concentrations of S^{2-} and SO_4^{2-} were considerably lower (at 0.5 and 2.1 mg L^{-1}, respectively), indicating that only 5% of the influent sulphur was converted into sulphide. In contrast, 4.62 mg L^{-1} of elemental sulphur was recovered from the UASB effluent, representing 46% of the sulphur load in the raw sewage, thereby confirming that micro-aeration applied above the reaction zone of the UASB reactor favoured the partial oxidation of sulphide species, as previously reported.[32] Under aerobic conditions, oxygen serves as an electron acceptor in the oxidation of sulphide by chemolithotrophic bacteria, [14,33,34] especially those of the genus *Thiobacillus* that display a great affinity for sulphur compounds. Under limiting oxygen conditions, the formation of S^0 is favoured, while increased oxygen concentration promotes the production of SO_4^{2-}, [35,36] thereby contributing to the undesirable complete oxidation of S^{2-} to SO_4^{2-}.

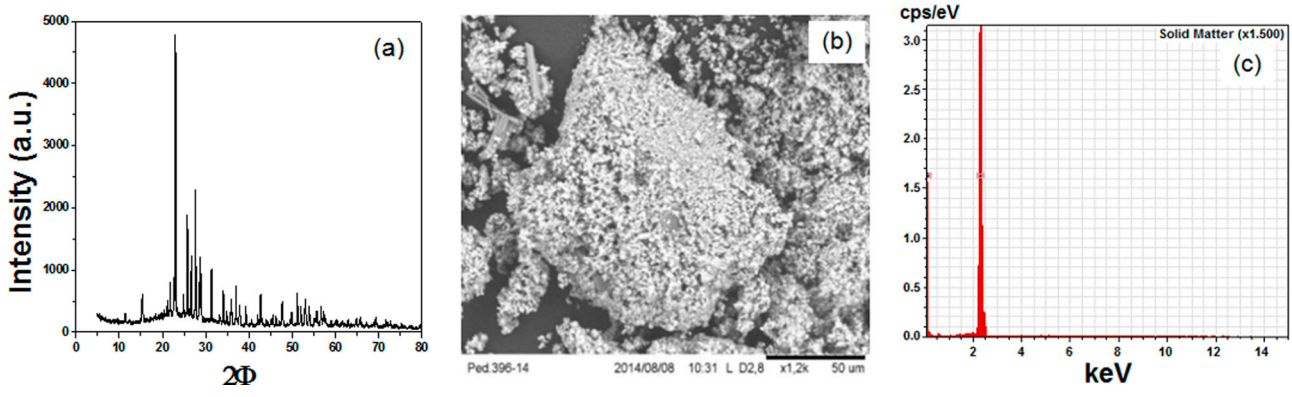


Figure 2. (**a**) XRD spectrum, (**b**) scanning electron micrograph and (**c**) energy dispersive X-ray spectrum of the insoluble material precipitated in the effluent from the filter of the AH reactor showing the spectral and microscopic characteristics of elemental sulphur.

3.3. Mass balance of sulphur in the micro-aerated UASB reactor

In order to understand the behaviour of sulphur species in the micro-aerated UASB reactor, the mean concentrations of sulphur forms with oxidation state +6 and –2 were assessed. The levels of dissolved SO_4^{2-} and S^{2-} in the liquid phase were determined in the influent (raw sewage), effluent and sludge according to Equations (2)–(4), and the results are presented in Figure 3. Of the total sulphur load present in the influent (502 mgS day^{-1}), 20.9% (105 mgS day^{-1}) of SO_4^{2-} and 4.98% (25 mgS day^{-1}) of S^{2-} remained dissolved in the effluent, while 46% (231 mgS day^{-1}) was recovered from the effluent as insoluble S^0. Considering that 11.12% of the total sulphur load was found in the sludge (55.85 mgS day^{-1}), it is likely that the residual 16.96% represents the volume of H_2S produced by the UASB reactor which, under the experimental conditions employed (30°C and 0.95 atm), would amount to 77 mL day^{-1}.

According to Speece,[37] it is very difficult to close the mass balance of sulphur for an anaerobic process owing to the problem of sulphide conversion, which may be caused by the microbial synthesis of sulphur resulting in the internal precipitation of sulphide produced. The discrepancy observed in the experimental data obtained in the present study can be explained by the different pathways of conversion of dissolved SO_4^{2-} and S^{2-} in the UASB reactor. For example, SO_4^{2-}can be reduced to $S^{2-}$for the formation of organic compounds through an assimilatory route or for the production of energy by a dissimilation pathway. In the latter case, the S^{2-} is excreted into the medium where it may precipitate through combining with a metal and accumulate in the sludge in the form of acid volatile sulphides [22,38] that can be used as a source of energy, [39] or be released in the gas phase (H_2S stripping). Another important consideration is that a portion of the sulphur, theoretically immobilized as bioparticles in the sludge, may actually be carried away in the effluent and, therefore, not measured. In a recent study of the mass balance of sulphur in a UASB reactor fed with raw sewage, Souza [40] reported that 15–18% of the original sulphur load was present as SO_4^{2-} in the effluent, 57–64% as S^{2-} in the effluent, 3–5% as S^{2-} precipitated in the retained sludge, 4–5% as S^{2-} absorbed by cells in the retained sludge, 2–4% as S^{2-} precipitated in the effluent sludge, 2–5% as S^{2-} absorbed by cells in the effluent sludge and 1% was present as S^{2-} in the biogas. Typical concentrations of methane and H_2S in biogas produced by UASB

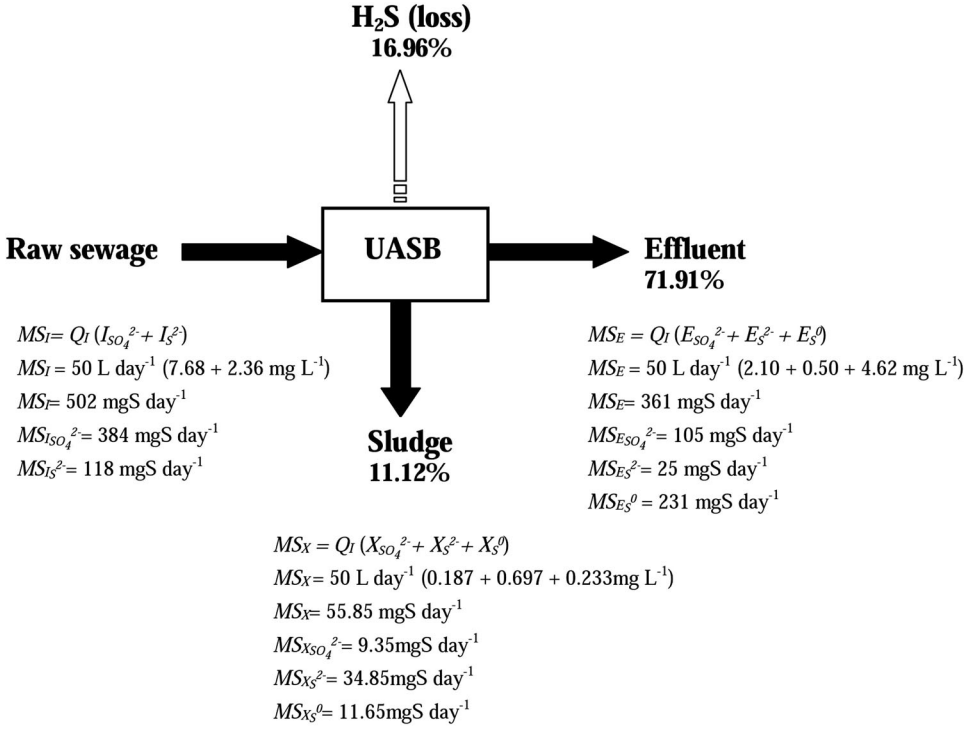


Figure 3. Mass balance of sulphur species in the micro-aerated UASB reactor. Notes: MS_I, MS_E, MS_X = total loads (mgS day^{-1}) of sulphur species in the influent, effluent and sludge, respectively; MS_L = loss (mgS day^{-1}) of sulphur in the form of H_2S; Q_I = flow rate of influent (L day^{-1}); SO_4^{2-}, S^{2-}, S^0 prefixed with I, E,X = levels (mgS L^{-1}).of sulphur species in the respective reactor fractions; $MS_{I_{SO_4^{2-}}}$, $MS_{I_S^{2-}}$, $MS_{E_{SO_4^{2-}}}$,$MS_{E_S^{2-}}$, $MS_{E_S^0}$, $MS_{X_{SO_4^{2-}}}$, $MS_{X_S^{2-}}$ and $MS_{X_S^0}$ = mass balances(mgS day^{-1}) of sulphur species in the respective reactor fractions.

reactors fed with domestic sewage are, respectively, 70–80% and 0.1–0.4%.[41,42]

3.4. Oxygen flow in the micro-aerated UASB reactor

In the absence of homogenization, the concentration of dissolved oxygen determined above the digestion zone oscillated between cycles and the values obtained were not reproducible. However, since the airflow was maintained at a mean value of 1.2 ± 0.39 L h^{-1}, and considering that the specific mass of air is 1.2 g L^{-1} while the oxygen fraction in air is 23.3% (w/w), the amount of oxygen supplied to the micro-aeration unit was in the region of 334 mg h^{-1} or 10.44 mmol h^{-1}. Micro-aeration during the trial period ensured a constant concentration of dissolved oxygen around 0.46 ± 0.088 mg L^{-1}, permitting the conversion of sulphur species into the most oxidized forms, such as SO_4^{2-} and S^0. The oxygen consumption rate above the aeration zone was 0.70 ± 0.14 mgO_2 L^{-1} h^{-1}, a value that was about 70 times lower than that of the conventional activated sludge process.[43] The flow rate of the raw sewage in the UASB reactor employed in the present study was 2.08 L h^{-1} and the total sulphur load was 10.04 mg L^{-1}, hence the sulphur flow rate was 20.88 mg h^{-1} or 0.65 mmol h^{-1} giving a mmolO_2/mmolS rate ratio of approximately16.0 mmol h^{-1}.Van der Zee et al. [18] obtained mmolO_2/mmolS ratios of between 8 and 10 mmol h^{-1} in a micro-aerated anaerobic reactor, and found that these levels were sufficient to ensure that the amount of sulphide in the biogas was reduced to concentrations of approximately 0.02%. Moreover, these authors reported that such molar ratios favoured the oxidation of S^{2-} to S^0, $S_2O_3^{2-}$ and, most likely, polysulphides, by routes that did not involve the re-oxidation of reduced sulphur species to SO_4^{2-}.

According to Jansen et al., [13] significant amounts of thiosulphate are formed when the O_2/S ratio in a reactor is below 0.7, for example, indicating that the chemical oxidation of S^{2-} to $S_2O_3^2$ is favoured under limiting oxygen conditions. In the present study, the biological oxidation of S^{2-} followed first-order reaction kinetics ($k = 0.28$–0.47 h^{-1}) during the 120 days of operation of the system. However, the chemical oxidation of S^{2-} could also be described as a first-order reaction ($k = 0.87$ h^{-1}) and it is likely, therefore, that the system would have a lower capacity to perform biological oxidations in comparison with chemical oxidations under limiting oxygen conditions. On the other hand, under limiting S^{2-} conditions, the oxidation of S^{2-} by *chemolithotrophic* Thiobacillus *is favoured in comparison with chemical oxidation* by virtue of the strong affinity of these microorganisms for sulphur.[36] The *complete oxidation of* S^{2-} to SO_4^{2-} by *colourless sulphur bacteria* generates more useful metabolic energy than partial oxidation. In order to obtain partial oxidation, it would be necessary to supply high sulphur loads or low oxygen concentrations.[44]

In the present study, the effects of micro-aeration on the removal of COD from the final effluent were not investigated although the O_2/COD ratio in the reactor was estimated to be 0.027 (cf. Table 3). In this context, Van der Zee et al. [18] reported that the removal of COD was insignificant when the O_2/COD ratio was around 0.05.

4. Conclusions

The AH reactor not only removed a considerable fraction of organic material (77%) but also produced a final effluent with a low concentration of S^{2-} (3.87% of the total sulphur load), resulting in a significant reduction in the unpleasant odour typical of anaerobic effluents. It was achieved due to a complete oxidation of S^{2-} to SO_4^{2-} (56.1% of the total sulphur load) carried out, most probably, by *anoxygenic purple and green bacteria*. The partial biological oxidation that occurred in the AH reactor allowed the recovery of 30% of the sulphur load as S^0. Regarding the micro-aerated UASB reactor, only 4.98% of the sulphur load was released in the effluent as S^{2-}, whereas 20.9% was present in the form of dissolved SO_4^{2-} and 46% was precipitated in the form of S^0. A small amount of S^0 (2.32%) was also found in the sludge. Thus, anaerobic filtration or micro-aeration carried out above the digestion zone of a UASB reactor favoured the biological oxidation of S^{2-} and the release of relatively odourless effluents. Both technologies represent feasible and low-cost alternatives for the anaerobic treatment of domestic sewage.

Disclosure statement

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